

Experimental evidence for partial conjunct agreement in Turkish

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ABSTRACT

In several languages, when a head agrees with a conjoined phrase, it may express either the sum of the features of all conjuncts (i.e., *full agreement*) or the features of only a subset of conjuncts (i.e., *partial agreement*). Investigations into (partial) agreement with conjunction have played a crucial role in linguistic theorizing of agreement. Here we aim to contribute to this literature by experimentally investigating verbal agreement strategies with coordinate and disjunctive subjects in Turkish. Using three acceptability judgment tasks, we test the availability of partial verbal agreement with preverbal and postverbal coordinate subjects and preverbal disjunctive subjects. For coordinate subjects, we find that full agreement is the most acceptable strategy, while partial agreement with the last (or second) conjunct 'I' is *somewhat* available in both word orders. For disjunctive subjects, acceptability is overall degraded, suggesting a grammatical gap in assigning number with disjunction. Nevertheless, similar to coordination, full agreement remains the most acceptable option, followed by partial agreement with the last conjunct.¹

1 Introduction

It is widely observed and empirically attested across several languages that the target of agreement can be only *one* or more of the conjuncts within a conjoined phrase (e.g., Koutsoudas 1968). That is, the agreement head may bear the features (e.g., ϕ , noun class) of only the second (as in 1a) or only the first conjunct (as in 1b), resulting in partial agreement. It is also possible for two agreement heads to bear the features of two different conjuncts (as in 1c). Partial agreement is often available alongside full (or resolved) agreement (i.e., agreement with the sum of the features of both/all conjuncts, as in 1a) or default agreement (i.e., morphological last resort when no feature value is provided) (see Nevins & Weissner 2019, for a review). The examples in 1a-1c illustrate partial or full agreement (features bolded) in Turkish, Standard Arabic and Polish, respectively.

- (1) a. [Sen-in-le ben-im] ara-m/mız-da kocaman bir
you-GEN-COM I-GEN between-1SG/1PL.POSS-LOC huge a
fark var.
difference exist
'There is a huge difference between you and me.'
(from the song *Fark Var* by Ceza)

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- b. Qaraʔat [ʕaliyaa wa ʕumar] l-qiṣṣa
read.3SG.FEM Alia.3SG.FEM and Omar.3SG.MASC the-story
'Alia and Omar read the story.'
(Aoun *et al.* 1994)
- c. Maria chce, żebym [ja i mój sąsiad]
Maria wants that.COND-1SG I and my neighbor.SG.MASC
wyszedł
left.SG.MASC
'Maria wants me and my neighbor to leave.'
(Citko 2018)

Structurally, coordinate phrases are often represented by a coordination head (&), whose specifier and complement positions are asymmetrically filled by the first and second conjuncts, respectively (e.g., Benmamoun *et al.* 2009, Munn 1993, cf. Haskell & MacDonald 2005). This is shown in 2.



A common theoretical assumption is that neither the coordination head (&) nor its maximal projection (&P) carries ϕ features (e.g., Citko 2018). Consequently, when an Agree head targets a coordinate phrase, its features cannot be valued by the phrase itself, potentially leading to non-standard agreement routes. Given the asymmetric structure in 2, partial agreement with the first conjunct, as in Example 1b, can be explained by the hierarchically prominent position of the first conjunct. However, partial agreement with the last, linearly closer conjunct of a preverbal phrase, as illustrated in Turkish (Example 1a), is particularly notable for syntactic theories of agreement. This is because agreement with the second conjunct requires the Agree head to access the second conjunct by virtue of its linear proximity rather than its hierarchical position or c-command relations. This challenge is heightened by the fact that languages exhibiting sensitivity to linear proximity in conjunct agreement often do not show the same sensitivity in other grammatical contexts.

Existing theoretical approaches offer contrasting views on whether linearity-sensitive partial agreement should be analyzed within the syntax proper or assigned to a post-syntactic component. One group of accounts derives partial agreement entirely in the syntax. Mechanisms that enable partial agreement in these accounts include the equidistant positioning of the coordinate phrase (&P) and the conjunct in its specifier relative to the Agree head (as argued for Dutch by van Koppen 2005), the presence or absence of movement operations in the language or the ordering of Agree and Merge operations (as proposed for Serbo-Croatian by Bošković 2009 and Murphy & Puškar 2018, respectively). However, a major limitation of these accounts is their lack of generalizability; the language-specific structural and/or operational mechanisms that they rely on often limit their ability to explain partial agreement cases across languages.

Another group of accounts adopts a two-stage agreement process that incorporates a post-syntactic component (e.g., Benmamoun *et al.* 2009, Bhatt & Walkow

2013, Köylü 2018, Marušič *et al.* 2015, Tat & Kornfilt 2022). In this line of research, after a relation is established between an Agree head and its target in the syntax, the features of one or more conjuncts are transferred to the head post-syntactically, (i.e., at the PF). Crucially, this second step may be sensitive to linear proximity such that the features of the conjunct that is linearly closer to the head may be copied, resulting in partial agreement.

Compared to the relatively extensive literature on agreement with coordinate phrases, previous work on agreement with disjunctive phrases (e.g., those using operators like ‘or’) is limited. Although coordinate and disjunctive phrases are often considered syntactically similar and both are assumed to lack inherent grammatical number, they differ semantically: Coordinate phrases typically denote semantic plurality (unless they refer to a single entity), whereas disjunctive phrases do not (e.g., Peterson 2004). This semantic difference may lead to a scenario where speakers’ decisions about (verb) number with disjunctive phrases are often ad hoc and non-deterministic due to the absence of a clear grammatical principle, creating what is sometimes termed *grammatical gap* (Foppolo & Staub 2020).

This view is supported by empirical findings. When the subject is a disjunction of two singular nouns, both English and Italian speakers were found to have degraded ratings for both singular and plural verb agreement, suggesting that neither form is fully acceptable (Foppolo & Staub 2020). Furthermore, English speakers had comparable acceptability and showed no evidence for disruption in eye-movements when encountering a verb in either form, providing further support that both forms may be available to a similar extent. As in coordinate phrases, agreement with disjunctive phrases is also subject to linear proximity effects (Haskell & MacDonald 2005, Keung & Staub 2018, Peterson 1986). For example, it has been shown for disjunctive subject phrases in English that the verb form that agrees with the linearly closer conjunct is selected or produced more often and rated higher than the one that agrees with the first conjunct (e.g., *Can you tell me whether the horses or the clock is/are red?*).

In this study we investigate the extent to which these cross-linguistic observations concerning verb agreement hold in Turkish. Specifically, we examine the availability of partial agreement on the Turkish verb with both coordinate and disjunctive subject phrases. In the remainder of the paper, we review previous work on conjunct agreement phenomena in Turkish in Section 2, present the methodology and results of three acceptability judgment tasks in Sections 3-5 and discuss the general findings and lay out future directions in Section 6.

2 Previous work on conjunct agreement in Turkish

In Turkish subject-verb and genitive-possessive relations exhibit agreement, bearing person and number features (Kornfilt 1997). While full agreement is the standard, partial agreement is reported to be an available strategy in both nominal and verbal domains. The canonical word order in Turkish is SOV.

Tat & Kornfilt (2022) present data, including those from corpora, to illustrate cases of partial agreement in nominal phrases and clauses in Turkish. They show that when the possessor noun in a DP or the embedded subject in a nominalized clause is a coordinate phrase, the possessee noun (as in 3a) or the embedded verb

(as in 3b) may have either full agreement (i.e., 1PL for 1&2SG; see İnce & Aydın 2015) or partial agreement with the last/second conjunct only (i.e., 1SG).

- (3) a. [sen-in ve ben-im] hayal-ler-imiz/im/*in
 you-GEN and I-GEN dream-PL-1PL/1SG/2SG.POSS
 ‘your and my dreams’
 b. [o-nun ve ben-im] oyuncuyu yakından
 s/he-GEN and I-GEN actor closely
 tanı-ma-mız/m/*sı
 know-NMLZ-1PL/1SG/3SG.POSS
 ‘her/his and my knowing the actor closely’

Partial agreement strategies have also been observed and experimentally attested for the verbal domain (Köylü 2018). Using a rating and a forced choice task, Köylü examined the acceptability of sentences with coordinate subjects, in which the verb either agreed with the sum of the features of both conjuncts or the first or the second conjunct only, i.e., 1PL, 3SG and 1SG in 4², respectively.

- (4) [O ve ben] git-ti-k/m/??∅.
 s/he and I go-PST-1PL/1SG/3SG
 ‘S/he and I went.’

The rating task revealed that sentences with full agreement received higher ratings than those with partial agreement, regardless of the subject’s linear position with respect to the verb, i.e., preverbal or noncanonical postverbal. Within partial agreement, sentences with agreement with the linearly closer conjunct (e.g., *ben* ‘I’ in 4) were rated as more acceptable than those with agreement with the linearly distant conjunct. This was further confirmed by a forced-choice task, in which Turkish speakers reliably preferred sentences with partial agreement with linearly closer conjuncts over those with partial agreement linearly distant conjuncts.

Both Tat & Kornfilt (2022) and Köylü (2018) propose a two-stage process for full/partial agreement, involving a syntactic and a post-syntactic component. However, their accounts differ with respect to the computation of ϕ features and how partial agreement is realized. For Köylü (2018), the syntactic stage involves an Agree relation between the agreement head and a conjunction phrase that inherits the features of both conjuncts, leading to full agreement. Partial agreement with the linearly closer conjunct, however, is a PF phenomenon that happens after spell-out. Tat & Kornfilt (2022) propose that in the syntactic stage, interpretable features of the goal are checked against uninterpretable features of the probe. The morphological realization of these features is a separate, post-syntactic process that happens after spell-out and before the phonological form is received. This morphological step is where the features can be altered during lexical insertion, leading to partial agreement.

Lastly, Paparounas & Akkuş (2024) report cases of default agreement on the verb of a nominalized embedded clause. They show that for some Turkish speakers,

²The example in 4 is constructed based on the descriptions in Köylü’s (2018) paper.

when the subject of a nominalized clause is a coordinate phrase including a local and a non-local person such as ‘I and Leyla’ in 5, the Agree probe of the genitive case $-(n)In$ cannot be valued, resulting in the insertion of the default 3SG agreement on the verb (see Paparounas & Akkuş 2024 for other, non-coordinated DPs that lead to default agreement).

- (5) Herkes [ben ile Leyla]-nın beraber gel-diğ-in-i
 everyone I and Leyla-GEN together come-NMLZ-3SG.POSS-ACC
 gör-dü.
 see-PST
 ‘Everyone saw that I and Leyla came together’.

Overall, while there is evidence for partial agreement in Turkish, the contexts in which it is attested and empirically tested remain limited. This study expands on prior research by using three acceptability judgment tasks to investigate agreement with coordinate and disjunctive subject phrases in the Turkish verbal domain. To anticipate the results, we find that Turkish speakers find full agreement resolving the features of both conjuncts more acceptable than partial agreement. This preference is less pronounced with disjunctive subjects, which are overall less acceptable than coordinate subjects. There is also evidence for the acceptability of partial agreement with the second conjunct for both coordinate and disjunctive subject phrases combining a proper name and a local pronoun (e.g., *Yıldız and I*).

3 Experiment 1: NP vs. clausal coordination accounts

Experiment 1 is designed to determine whether partial agreement is a genuine agreement strategy in Turkish, or if it is a byproduct of other syntactic processes involving regular agreement strategies. It has been argued that partial agreement may result from an underlying structure in which constituents larger than NPs, such as clauses, are coordinated (Aoun *et al.* 1994). This structure would then undergo Right Node Raising and/or ellipsis, which makes it appear as if the verb only agrees with one of the conjuncts. This is illustrated with a simplified configuration in 6.

- (6) [[_{IP} NP₁ ~~Verb~~] and [_{IP} NP₂ Verb]]

Previous experimental work on Turkish provided evidence for partial agreement with the linearly closer conjunct in Turkish (Köylü 2018), but all the verbs in the stimuli were distributive verbs that allow for a clausal coordination reading. In Experiment 1, we investigate whether partial agreement is possible with collective verbs such as *gather* which do not allow for clausal coordination. This experiment was preregistered at: <https://osf.io/kxb6a/overview>

3.1 Participants

69 native speakers of Turkish ($M_{age} = 24.5$) participated in the task. Participants (including those in Experiments 2 & 3) were recruited through social media. Each participant was compensated 100 Turkish Lira (~\$3.5 at the time of the study).

3.2 Materials

The study had a 2x3 design, crossing PREDICATE (i.e., distributive such as *telaşlan-* ‘to worry’ or collective such as *toplan-* ‘to gather’), and AGREEMENT on the verb (i.e., Full Agreement (FA) bearing 1PL, Last Conjunct Agreement (LCA) bearing 1SG or First Conjunct Agreement (FCA) bearing the morphophonologically null 3SG). All stimuli had preverbal subjects including a proper name followed by the 1SG pronoun coordinated with *ve* ‘and’ as in *Yıldız ve ben* ‘Yıldız and I’. There was an adverbial phrase before the verb to reveal whether partial agreement would arise when the subject was not adjacent to the verb. An example set of stimuli is in 7.

- (7) [Yıldız ve ben] görüşme için telaşlan-/toplan-dı-k/m/∅=FA/LCA/FCA.
Yıldız and I meeting for worry-/gather-PST-1PL/1SG/3SG
‘Yıldız and I {worried about, gathered for} the meeting.’

Six experimental lists were created in a Latin-square design. Each list included 36 target items and 72 fillers. 24 of the fillers were grammatical, 24 of them were ungrammatical, and the last 24 had coordinate objects.

3.3 Procedure

Experiment 1 (and Experiments 2 & 3) was conducted online on PCIBex. In the task, each sentence was presented as a whole, and the participants were instructed to rate it on a scale of 1-7, where 7 indicated ‘definitely a sentence that I might produce or hear from Turkish speakers’ and 1 ‘definitely not a sentence that I might produce or hear from Turkish speakers’. With instructions emphasizing daily speech, we aimed to tap into the acceptability of sentences in daily speech rather than their acceptability in written forms per prescriptive rules. Each sentence was followed by a comprehension question. There were three practice trials, for which feedback was provided.

3.4 Data analyses

The data were analyzed with mixed effects models (Bates *et al.* 2015) using the CLMM function in the `ordinal` package in R (Christensen 2018). The dependent variable was acceptability ratings. Fixed effects were PREDICATE and AGREEMENT, with participants and items as random effects. PREDICATE was coded using sum contrasts. AGREEMENT was coded using Helmert contrasts, which compared the effects of FULL (FA vs. LCA & FCA) and LAST CONJUNCT AGREEMENT (LCA vs. FCA). We fit two models: a crossed and a nested model. The crossed model had both fixed effects and their interaction. The nested model had a main effect of PREDICATE and separate AGREEMENT contrasts within distributive and collective predicate conditions. Each model (in Experiments 1-3) included random intercepts for participants and items. Absolute *z*-scores that are 2 or above indicate significance at the α level .05.

3.5 Results

Before the main analyses, we first inspected participants’ accuracy in the filler items. All participants had a mean accuracy above 70%, so they were all included in the analyses. The distribution of participants’ ratings is given in Figure 1.

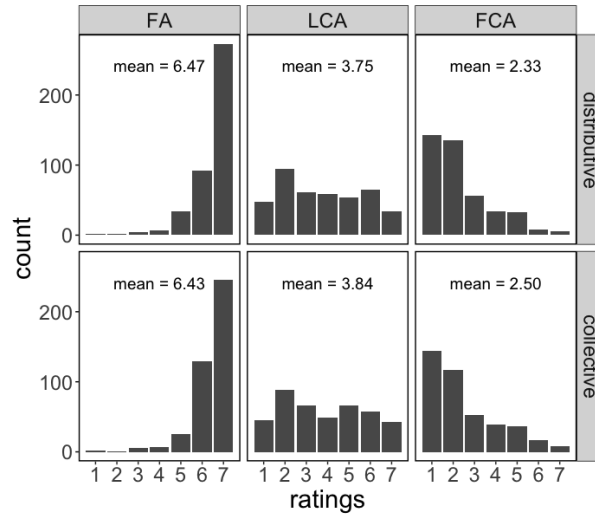


Figure 1: Distribution of ratings (7 = ‘definitely acceptable’) in Experiment 1, showing the count of responses per rating and mean ratings per condition. FA, LCA and FCA stand for Full, Last Conjoint and First Conjoint Agreement, respectively.

The results of the statistical analyses³ showed that overall, sentences with full agreement were more acceptable than those with partial agreement, i.e., FA vs. LCA & FCA ($\hat{\beta} = 5.01$, $SE = 0.14$, $z = 37.08$). Within partial agreement strategies, sentences with LCA were more acceptable than those with FCA ($\hat{\beta} = 1.74$, $SE = 0.10$, $z = 18.35$), leading to the FA > LCA > FCA pattern in acceptability. There was an interaction between PREDICATE and FULL AGREEMENT ($\hat{\beta} = 0.41$, $SE = 0.18$, $z = 2.30$). Pairwise comparisons revealed the nature of this interaction: While the FA > LCA > FCA pattern holds for both distributive and collective verbs, the difference between full and partial agreement was slightly larger for distributive verbs ($\hat{\beta}$ ’s ≥ 1.72 , $SE \geq 0.13$, $z \geq 13.19$). There were no other effects (z ’s ≤ 0.22).

To further investigate the availability of FCA as an agreement strategy, we compared the ratings for FCA sentences and ungrammatical fillers. A paired-samples *t*-test showed no evidence for a difference between FCA ($M = 2.41$, $SE = 0.15$) and ungrammatical sentences ($M = 2.43$, $SE = 0.09$), $t(68) = 0.16$, $p = .87$.

3.6 Discussion

Experiment 1 showed that overall, sentences with full agreement received the highest ratings. Within the two partial agreement strategies, sentences with agreement with the linearly closer conjunct received higher ratings than those with agreement with the linearly distant conjunct, yielding the FA > LCA > FCA pattern. Crucially, this was the case with both distributive and collective verbs, with smaller differences and slightly better ratings for partial agreement with collective verbs. These findings suggest that LCA as an agreement strategy cannot be merely due to a clausal coordination reading, but rather it is available in the case of strict NP-coordination. A limitation of Experiment 1, however, is that the low ratings observed for FCA

³Output from all statistical analyses is available at: <https://osf.io/c2pqh/files>

may be due to the use of the morphophonologically null 3SG marking. We address this potential confound in Experiment 2.

4 Experiment 2: The role of conjunct type and conjunct order

Experiment 1 provided evidence for partial agreement with the second conjunct in canonically positioned, preverbal coordinate subjects of the form [Proper Name + 1SG pronoun] (e.g., [*Yıldız and I*]). Experiment 2 tests the generalizability of this effect across three key dimensions. First, the availability of partial agreement with a 1SG pronoun may stem from its unique pronominal status within the coordinate phrase. Experiment 2 addresses this by comparing coordinate subjects with either one or two pronouns as in [Name + Pronoun] (a proper name and a 1SG pronoun) and [Pronoun + Pronoun] (composed of 1SG and 2SG pronouns). Second, partial agreement cases in Experiment 1 may be attributed to the surface linear proximity of the last conjunct ‘I’ to the Agree probe. Experiment 2 tests whether partial agreement is available in the absence of linear proximity by placing the 1SG pronoun as the linearly distant conjunct from the verb. Finally, to test whether partial agreement is restricted to preverbal subjects in canonical word order, Experiment 2 tests coordinate subjects in both preverbal and noncanonical postverbal positions. This experiment was preregistered at: <https://osf.io/kvw8f/overview>

4.1 Participants

94 native speakers of Turkish ($M_{age} = 25.2$) participated in the task. Each participant was compensated 120 Turkish Lira (~\$3.5 at the time of the study).

4.2 Materials

The study had a 2x2x3 design, crossing ORDER (i.e., preverbal or postverbal subjects), CONJUNCT (i.e., coordinate subjects combining the 1SG and 2SG pronouns or the 1SG pronoun and a proper name), and AGREEMENT on the verb as in the previous experiment (i.e., FA, LCA with the second conjunct or FCA with the first conjunct). All the stimuli had subjects coordinated with *ve* ‘and’. Differently from Experiment 1, the preverbal subject conditions had the 1SG pronoun as the initial conjunct in the subject phrase, making it the conjunct that is linearly more distant from the verb. To keep the linear distance between the 1SG pronoun and the verb constant across conditions, the order of the conjuncts was reversed in postverbal subjects, placing the 1SG pronoun as the second/last conjunct. Unlike Experiment 1, Experiment 2 tested FCA sentences featuring an overt morphological marking to establish whether the availability of FCA depends on the overt realization of agreement. An example set of stimuli is in 8.

- (8) a. [Ben ve Yıldız/sen] uzun zaman bekle-di-k/∅/n/m.
I and Yıldız/you long time wait-PST-1PL/3SG/2SG/1SG
‘I and {Yıldız, you} waited for a long time.’
b. Bekle-di-k/m/∅/n uzun zaman [Yıldız/sen ve ben].
wait-PST-1PL/1SG/3SG/2SG long time Yıldız/you and I.
‘{Yıldız, you} and I waited for a long time.’

Twelve experimental lists were created in a Latin-square design, each containing 48 target items and the same 72 filler items from Experiment 1.

4.3 Data analyses

We ran a crossed model and a nested model using the fixed effects of ORDER, CONJUNCT and AGREEMENT. The crossed model included all three fixed effects and their two- and three-way interactions. The nested model had main effects of ORDER and CONJUNCT and separate AGREEMENT contrasts within each unique combination of CONJUNCT and ORDER, i.e., preverbal subjects with two pronouns, preverbal subjects with a proper name and the same for postverbal subjects. The investigation of agreement contrasts within each such combination allowed us to closely examine how the acceptability of agreement patterns changed in relation to conjunct properties such as order and pronominal status. ORDER and CONJUNCT were sum-contrast coded, and AGREEMENT was coded using Helmert contrasts as in Experiment 1. The rest of the analysis steps were the same as in Experiment 1.

4.4 Results

Four participants' accuracy was below 70% in the filler items; hence, they were excluded from the analyses. The mean accuracy of the remaining 90 participants was 94.4%. The distribution of participants' ratings is given in Figure 2.

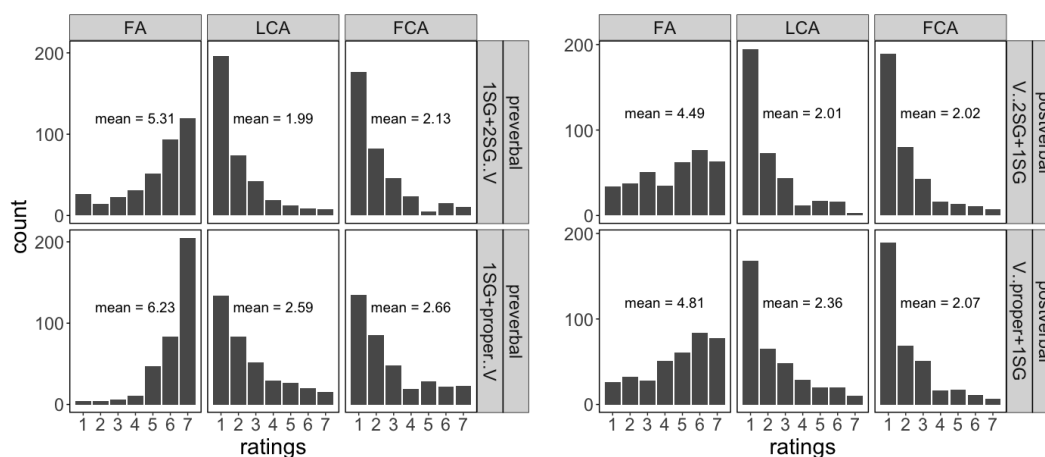


Figure 2: Distribution of ratings (7 = 'definitely acceptable') in Experiment 2, showing the count of responses per rating and mean ratings per condition. Preverbal conjuncts are on the left; postverbal conjuncts are on the right.

The results of the statistical analyses showed that overall, sentences with a proper name and those with preverbal subjects were more acceptable than those with two pronouns and those with postverbal subjects, respectively ($\hat{\beta}$'s ≥ 0.68 , SE 's ≥ 0.06 , z 's ≥ 11.36). For agreement, sentences with full agreement were overall more acceptable than those with partial agreement ($\hat{\beta} = 4.10$, $SE = 0.09$, $z = 46.06$). Within partial agreement, there was no evidence for an overall difference between LCA and FCA ($\hat{\beta} = 0.01$, $SE = 0.08$, $z = 0.11$). All two-way interactions

between the predictors were significant ($\hat{\beta}$'s ≥ 0.29 , SE 's ≥ 0.15 , z 's ≥ 2.01); since our focus is on agreement patterns within unique subsets of data combining different conjunct types and orders, we did not examine those interactions further.

There was a trend for a three-way interaction between ORDER, CONJUNCT and FULL AGREEMENT ($\hat{\beta} = 0.50$, $SE = 0.26$, $z = 1.93$), but it did not reach significance. Pairwise comparisons within combinations of different conjunct types and order revealed that sentences with full agreement were more acceptable than those with partial agreement in all cases ($\hat{\beta}$'s ≥ 3.42 , SE 's ≥ 0.14 , z 's ≥ 25.34). When the subject was postverbal and included a proper name, sentences with LCA were more acceptable than those with FCA ($\hat{\beta} = 0.40$, $SE = 0.14$, $z = 2.75$). There were no other effects (z 's ≤ 1.74).

4.5 Discussion

Confirming Experiment 1, the findings of Experiment 2 showed evidence for increased acceptability of full agreement over partial agreement across all tested configurations. A novel finding from Experiment 2 was that for coordinate phrases composed of the 1SG and 2SG pronouns, both partial agreement strategies (i.e., LCA and FCA) received similarly low ratings, suggesting no evidence for selective acceptability of LCA. Conversely, when the subject included a proper name followed by the 1SG pronoun in the second conjunct position in a postverbal environment, there was evidence for partial agreement with the pronoun. Given the surface order in this configuration (i.e., *V [Yıldız and I]*), the 1SG pronoun was the linearly distant conjunct from the verb. Crucially, there was no evidence for partial agreement with the same pronoun when it was the first conjunct in a preverbal subject phrase (i.e., *[I and Yıldız] V*). Taken together with the findings from Experiment 1, these results suggest that LCA as a partial agreement strategy arises when the 1SG pronoun is the only pronoun and appears as the second conjunct in a coordinate subject phrase, and this is so regardless of its proximity to the verb in the surface linear order. We leave the discussion of this finding to General Discussion.

5 Experiment 3: Agreement with disjunctive phrases

The goal of Experiment 3 is to reveal the verbal agreement strategies with disjunctive phrases in subject position including *veya* 'or' in Turkish, and to examine to what extent these strategies are similar to or different from those with coordinate phrases. This experiment was not preregistered.

5.1 Participants

93 native speakers of Turkish ($M_{age} = 23.5$) participated in the task. Each participant was compensated 120 Turkish Lira ($\sim \$3.5$ at the time of the study).

5.2 Materials

The study had a 2x2x3 design, crossing CONJUNCTION (i.e., coordination with *ve* 'and' or disjunction with *veya* 'or'), PREDICATE (i.e., verbal such as *gel-* 'to come' or nonverbal such as *hazır-* 'to be ready'), and AGREEMENT on the verb as in the

previous experiments (i.e., FA, LCA with the second conjunct or FCA with the first conjunct). All the stimuli had preverbal subjects including a proper name followed by the 1SG pronoun, as in Experiment 1. An example set of stimuli is in 9.

- (9) [Yıldız ve/veya ben] kahve için gel-/hazır-dI-k/m/∅.
 Yıldız and/or I coffee for come-/be.ready-PST-1PL/1SG/3SG
 ‘Yıldız {and, or} I {came, were ready} for the coffee.’

Twelve experimental lists were created in a Latin-square design, each containing 48 target items and the same 72 filler items from Experiment 1.

5.3 Data analyses

We ran a crossed model and a nested model using the fixed effects of CONJUNCTION and AGREEMENT. Since PREDICATE did not modulate the critical agreement patterns, the ratings for verbal and nonverbal predicates were aggregated (see below for details). The crossed model included CONJUNCTION and AGREEMENT as fixed effects and their interaction. The nested model had a main effect of CONJUNCTION and separate AGREEMENT contrasts within coordination and disjunction conditions. CONJUNCTION was sum-contrast coded, and AGREEMENT was coded using Helmert contrasts as in Experiments 1 and 2. The rest of the analyses were the same as before.

5.4 Results

Two participants’ accuracy was below 70% in the fillers, so they were excluded from the analyses. The mean accuracy of the remaining 91 participants was 96.3%.

The most complex model with all three predictors showed significant two-way interactions between CONJUNCTION and AGREEMENT (z ’s ≥ 6.90), but there was no evidence for a three-way interaction between PREDICATE and the other two predictors (z ’s ≤ 0.71). Since PREDICATE type did not modulate the critical interactions, the ratings for verbal and nonverbal predicates were aggregated. The distribution of participants’ ratings after this aggregation is in Figure 3. Subsequent analyses were run with the fixed effects of CONJUNCTION and AGREEMENT only.

The results revealed that overall, sentences with coordination were more acceptable than those with disjunction ($\hat{\beta}$ ’s ≥ 1.80 , SE ’s ≥ 0.06 , z ’s ≥ 28.11). For agreement, sentences with full agreement were more acceptable than those with partial agreement ($\hat{\beta} = 3.26$, $SE = 0.08$, $z = 40.74$). Within partial agreement, sentences with LCA were more acceptable than those with FCA ($\hat{\beta} = 1.60$, $SE = 0.07$, $z = 22.40$), replicating the FA > LCA > FCA pattern. There were also interactions between CONJUNCTION and AGREEMENT ($\hat{\beta}$ ’s ≥ 0.93 , SE ’s ≥ 0.14 , z ’s ≥ 6.86). While the FA > LCA > FCA pattern held for sentences with both coordination and disjunction, the differences were larger for coordination ($\hat{\beta}$ ’s ≥ 1.13 , SE ’s ≥ 0.10 , z ’s ≥ 11.62).

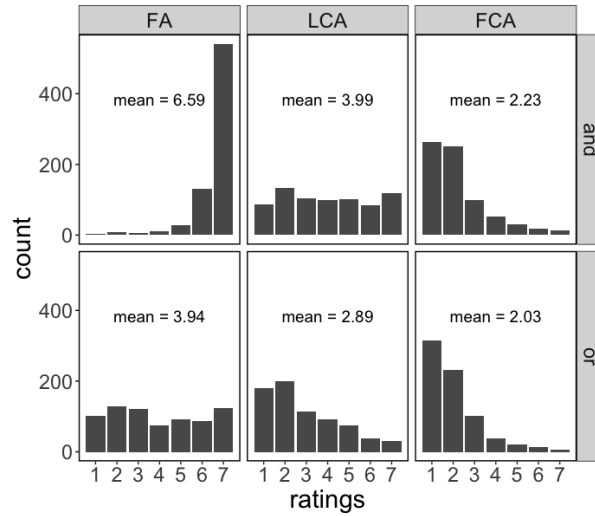


Figure 3: Distribution of ratings (7 = ‘definitely acceptable’) in Experiment 3, showing the count of responses per rating and mean ratings per condition. The ratings for PREDICATE were aggregated, in line with the analyses.

5.5 Discussion

Experiment 3 provided a direct comparison of verbal agreement strategies with coordinate versus disjunctive phrases in preverbal subject position. The results revealed that compared to sentences with coordinate phrases with *ve* ‘and’, the ratings for sentences with disjunctive phrases with *veya* ‘or’ were overall degraded. This was the case with both full and partial agreement. This provides novel evidence from Turkish supporting the hypothesis that no single form of agreement is fully acceptable with disjunction. Despite the low ratings, the FA > LCA > FCA pattern observed with coordination extended to disjunction, though the differences among agreement strategies were smaller than those for conjunction.

6 General Discussion

This study reported three acceptability judgment tasks that investigated verbal agreement strategies in Turkish for coordinate and disjunctive subject phrases. For coordinate subjects, the most acceptable strategy was full agreement, where the verb exhibits the sum of the features of both conjuncts (i.e., 1PL for 1&2/3SG). Partial agreement with the 1SG pronoun was also *somewhat* available with both preverbal and postverbal subjects. Crucially, partial agreement with 1SG emerged only when it served as the sole pronoun in the subject phrase and occupied the second conjunct position (e.g., *Yıldız and I*). That is, partial agreement was unavailable when the 1SG pronoun appeared as the first conjunct (e.g., *I and Yıldız*) or when the subject phrase contained a second pronominal competitor (e.g., *you and I*). For disjunctive subjects (which were tested only preverbally), no agreement strategy was fully acceptable. Nonetheless, similar to coordinate subjects, full agreement remained the most acceptable option, followed by limited acceptability of partial agreement with the 1SG pronoun in the last conjunct position (e.g., *Yıldız or I*).

An initial goal of the current study was to establish whether partial agreement in Turkish functions as a genuine agreement strategy or if it is merely a result of a clausal coordination (Aoun *et al.* 1994), whereby the verb agrees with a single conjunct in its clause, but surfaces as the verb form for the whole subject phrase after operations such as ellipsis or Right Node Raising. To test this, Experiment 1 employed collective verbs, which only allow nominal coordination and eliminate a clausal coordination possibility. The results showed that partial agreement with the second conjunct, i.e., last conjunct agreement or LCA, was available with both collective and distributive verbs. Moreover, the differences in acceptability between full and partial agreement were less pronounced for collective than distributive verbs—a pattern that directly opposes the predictions of a clausal coordination account. These findings provide compelling evidence against reducing partial agreement to clausal coordination, but rather provide support that partial agreement is available as a genuine grammatical strategy in Turkish (see Arsenijević *et al.* 2020 for similar empirical findings in South Slavic languages).

For partial agreement with coordination, the only phrase among the ones tested in Experiments 1 and 2 that enabled partial agreement was the one with a proper name followed by the 1SG pronoun (i.e., ‘[Yıldız and I] V’ or ‘V [Yıldız and I]’) (cf. Köylü 2018). Crucially, partial agreement with the pronoun in this configuration was available for both preverbal and postverbal subjects. While the 1SG pronoun is linearly adjacent to the verb in the canonical preverbal position, it is the linearly distant one in postverbal configurations. This outcome cannot be straightforwardly derived from hierarchical structure, given that the second conjunct occupies a lower structural position within the coordinate phrase. It is likewise difficult to reconcile this finding with surface-linear accounts of partial agreement (e.g., Bhatt & Walkow 2013, Tat & Kornfilt 2022), which attribute partial conjunct agreement to proximity between the agreeing conjunct and the verb in the surface linear order. While we cannot offer a comprehensive analysis of this finding due to the limited range of configurations tested here, we discuss several plausible explanations and outline directions for further research below.

At first glance, the availability of partial agreement with the 1SG pronoun in the second conjunct position might seem to suggest that partial agreement is available with the conjunct in the *linearly second position* within the coordinate phrase. This interpretation arises because the Agree probe can access the 1SG pronoun when it is the second conjunct, regardless of whether the entire coordinate subject phrase appears preverbally or postverbally (e.g., [Yıldız and I] V and V [Yıldız and I]), but it fails to do so when the pronoun occupies the first conjunct position in a preverbal subject (e.g., [I and Yıldız] V).

On theoretical grounds, however, it seems unlikely that partial agreement is conditioned on a conjunct’s linear position internal to the coordinate phrase. Instead, a more plausible alternative that unifies the preverbal and postverbal patterns is that Agree is established based on the conjunct’s proximity to the verbal probe in the canonical word order (where the second conjunct is linearly closer to the verb), while postverbal placement of the subject phrase is due to subsequent scrambling operations that apply after Agree is established. However, we refrain from drawing this conclusion at the moment given the limitations of our current data. To disentangle the roles of the conjunct’s position within the phrase and its linear proximity

to the verb, future investigations could, for example, test whether partial agreement is available with coordinate subjects in which the pronoun is the first conjunct but linearly closer to the verb (e.g., *V [I and Yıldız]*).

Moreover, to determine whether the observed partial agreement extends beyond the 1SG pronoun or is instead due to the prominence of first-person features on a person hierarchy (Harley & Ritter 2002), future studies should examine coordinate phrases excluding the 1SG pronoun (e.g., *Yıldız and you/she*). This would also clarify the lack of partial agreement with coordinate subjects involving 1SG and 2SG pronouns in Experiment 2 of the current study, revealing whether this is due to feature competition or a broader structural constraint on pronominal coordination.

Turning to disjunctive phrases, our results from Experiment 3 provided novel empirical evidence from Turkish for the absence of a fully acceptable agreement form with disjunction, converging with findings from English and Italian (Foppolo & Staub 2020, Keung & Staub 2018, Haskell & MacDonald 2005). We interpret the consistently low ratings with disjunction as supporting the hypothesis that grammar lacks a deterministic mechanism for resolving (verbal) number with disjunctive subjects (Foppolo & Staub 2020). This gap may force speakers to rely on ad hoc, probabilistic strategies for agreement. The reliance on such non-deterministic strategies may introduce a processing cost due to metalinguistic awareness, leading to the observed degraded ratings.

Despite the overall degraded ratings, Turkish speakers found full agreement more acceptable than partial agreement for disjunction, consistent with Italian production data (Foppolo & Staub 2020), though this preference was less pronounced than in coordination. Partial agreement with the 1SG pronoun, i.e., the linearly closer conjunct (e.g., *[Yıldız or I] V*), was the next most acceptable strategy. These findings demonstrate that the FA > LCA > FCA pattern established for preverbal coordination extends to disjunction in Turkish. Nonetheless, we do not conclude that all partial agreement constraints and patterns observed for coordination extend to disjunction, as Experiment 3 tested only a single type of subject phrase in the preverbal position. A comprehensive comparison between coordination and disjunction will require systematically testing disjunctive subjects in postverbal positions and across a broader array of conjunct combinations.

While the current study establishes that partial agreement is a genuinely available agreement strategy in Turkish, further work is necessary to fully uncover its empirical distribution and grammatical constraints. One important consideration is the extent to which partial agreement varies across and within individual speakers. For instance, Marušič *et al.*'s (2015) experimental work on gender agreement with coordinate subjects in Slovenian documented notable inter- and intra-speaker variation, suggesting evidence for the coexistence of multiple agreement grammars. To determine whether a similar profile exists for Turkish speakers, fine-grained statistical methods, such as Bayesian analyses that quantify individual-level variation, could be applied to the current dataset. Lastly, the overarching goal of this project is to formulate a comprehensive conjunct agreement system for Turkish. While the present experiments are a crucial step towards that goal, future investigations should test additional combinations of conjuncts varying in person, pronominal status and linear order, to reveal the full extent of conjunct agreement.

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